ClimatePulse: Temporal Fusion Transformers and Spatio-Temporal Graph Networks in Climate Modeling

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Abstract

Global climatic phenomena, regional pollution dispersion, and atmospheric field mechanics do not operate in isolation. They form a coupled, multi-scale physical system that no single modelling tradition has adequately captured. ClimatePulse binds four complementary components into one reproducible pipeline: (1) multi-horizon El Niño–Southern Oscillation (ENSO) forecasting via Temporal Fusion Transformers (TFT) with calibrated quantile regression; (2) PM2.5 dispersion modelling across 75 global monitoring stations through a 6,984-parameter A3TGCN Spatio-Temporal Graph Neural Network; (3) global weather-field prediction at 5.625° resolution using a 128-dimensional Encoder-Processor-Decoder Neural Weather Model (NWM) on a 2,000-node Fibonacci spherical mesh, conditioned on the Niño 3.4 index; and (4) Bayesian piecewise regression paired with CUSUM-based change-point detection for climate-feedback analysis.

On held-out test data, evaluation yields RMSE = 0.3505 °C and anomaly correlation ACC = 0.9449 for the TFT, translating to a skill score of 0.9051 against a persistence baseline of 0.1737, a 524.2 % relative improvement over the bidirectional LSTM baseline (skill score 0.1450). For PM2.5 exceedance detection at AQI > 100, the pollution ST-GNN records F1 = 0.9750 (precision 0.9677, recall 0.9825). NWM training converges to a best validation MSE of 0.1182 at epoch 78 of 89, with final training MSE of 0.1035. Across 918 months of GISTEMP temperature anomalies, Bayesian analysis detects no statistically reliable structural break; within the resolution of the data, the warming trend appears continuous and monotonic.

Keywords: Climate forecasting, Temporal Fusion Transformer, Spatio-temporal graph neural networks, Neural Weather Model, Encoder-Processor-Decoder, El Niño–Southern Oscillation, PM2.5 exceedance, Uncertainty quantification, Bayesian change-point detection

Research Article Format

1. Introduction

1.1 Background and Motivation

Few scientific problems carry the practical urgency of climate change, and fewer still demand the simultaneous command of ocean–atmospheric physics, environmental sensor networks, and probabilistic forecasting. Ocean-atmospheric anomalies — the El Niño–Southern Oscillation (ENSO) chief among them — ripple outward into precipitation regimes, temperature extremes, hurricane frequency, agricultural yields, and regional economic output. Anthropogenic emissions compound the picture. The same industrial processes that drive long-term radiative forcing also generate acute, localised air-pollution crises. Feedback loops between the two operate on timescales ranging from days to decades. Traditional siloed modelling frameworks struggle to capture these interactions.

1.2 Challenges in Existing Approaches

Three structural limitations recur across the literature.

- Numerical Weather Prediction (NWP) models, while physically grounded, carry computational latencies that render real-time decision support impractical.

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- Data-driven alternatives rarely embed uncertainty quantification in a principled, probabilistic manner. Point forecasts dominate; calibrated prediction intervals are an afterthought.
- Regional pollution-dispersion models tend to operate independently of the global atmospheric anomaly predictors — ENSO phase, sea-surface temperature gradients — that modulate stagnation events and long-range transport.

1.3 Research Gap

Separate advances in deep-learning weather forecasting and regional air-quality modelling have been substantial. High-fidelity Earth System Models (ESMs) such as CESM2 [15] preserve physical fidelity but at computational cost. Data-driven approaches like GraphCast [16] and FourCastNet [17] match ESM accuracy at roughly 1000× the speed, yet each remains a single-task system. What the field lacks is a unified, open-source pipeline. Multi-horizon ocean-atmospheric forecasting, sensor-graph air-quality prediction, spherical-mesh neural weather modelling, and statistical change-point analysis have never been coupled into a single reproducible workflow. ClimatePulse occupies that integration gap directly.

1.4 Research Questions

Three questions anchor the investigation.

- **RQ1.** Can attention-based Temporal Fusion Transformers with quantile regression outperform bidirectional LSTM baselines on multi-horizon ENSO forecasting, and what is the lead-time degradation profile?
- **RQ2.** Can A3TGCN Spatio-Temporal Graph Neural Networks reliably detect PM2.5 exceedance events with high precision and recall on a real-world global monitoring network?
- **RQ3.** Can an Encoder-Processor-Decoder ST-GNN trained on ERA5 reanalysis data learn global atmospheric mechanics on a Fibonacci spherical mesh without overfitting?

1.5 Key Contributions

Rather than a catalogue of isolated components, ClimatePulse offers a tightly coupled research platform. Four contributions follow from the design.

A) **Framework integration.** An open-source research platform binds TFT-based ENSO forecasting, A3TGCN pollution ST-GNN, an Encoder-Processor-Decoder Neural Weather Model on a Fibonacci spherical mesh, and Bayesian change-point detection into a single reproducible pipeline. Each module shares data infrastructure and evaluation tooling; none operates as a standalone script.

B) **Rigorous empirical validation.** Evaluation proceeds against persistence and LSTM baselines for ENSO; confusion-matrix and F1 metrics for pollution exceedance; and MSE convergence analysis across 89 NWM training epochs (0–88). Metrics are generated automatically from the pipeline, not hand-assembled.

C) **Spherical-aware neural weather modelling.** The NWM relies on a 2,000-node Fibonacci lattice mesh with grid-to-mesh, mesh-to-mesh, and mesh-to-grid bipartite GraphSAGE convolution stages. The architecture conditions on the Niño 3.4 index at every processor layer. Spherical-awareness is structural, not post-hoc.

D) **Reproducibility.** All data download, training, and evaluation steps are encapsulated in a declarative pipeline (`download-data`, `train`, `gen-eval-artifacts`), producing all paper figures and metrics without manual intervention.

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2. Related Work

2.1 Deep Learning for ENSO Forecasting

The attention mechanism introduced by Vaswani et al. [4] underpins the temporal self-attention layer at the heart of the TFT architecture [1]. Lim et al. [1] combine multi-head self-attention with LSTM encoder-decoder layers and gated variable-selection networks, yielding interpretable multi-horizon forecasts. Ham et al. [7] pushed ENSO prediction to 18-month lead times using CNNs trained on climate-model output, though their framework omits calibrated probabilistic outputs. The LSTM of Hochreiter and Schmidhuber [5] remains the principal recurrent baseline; Barnston et al. [6] established the skill-score evaluation standard against persistence that we adopt here.

Where Ham et al. [7] stop at point forecasts, the present work introduces quantile regression for calibrated uncertainty bounds and extends the comparison to include LSTM and persistence baselines across all 12 forecast months simultaneously.

2.2 Graph Neural Networks in Air-Quality Modelling

Non-Euclidean sensor networks resist regular-grid convolution, which explains why spatio-temporal graph neural networks have become the default architecture for air-quality forecasting. A3TGCN, introduced by Bai et al. [2], couples spectral graph convolution with GRU temporal evolution and sequence-level soft attention. The GCN spectral convolution foundation traces to Kipf and Welling [8]; diffusion convolutional networks for traffic forecasting (Li et al. [9]) share structural properties with atmospheric dispersion modelling, since both involve advection-diffusion over irregular topologies.

2.3 Neural Weather Modelling

GraphCast [16] and Keisler [23] pioneered the Encoder-Processor-Decoder approach on icosahedral meshes. FourCastNet [17] takes a different route, deploying Adaptive Fourier Neural Operators on regular grids. The NWM presented here adapts the Encoder-Processor-Decoder architecture to a Fibonacci-lattice spherical mesh with k-NN bipartite connectivity and adds ENSO conditioning through an additive MLP mechanism injected at every mesh-processor layer — a coupling absent from prior spherical GNN weather models.

2.4 Uncertainty Quantification in Climate Models

Proper scoring rules for probabilistic forecast evaluation were formalised by Gneiting and Raftery [11]. Quantile regression supplies distribution-free probabilistic forecasts [1], while Bayesian methods [12] yield full posterior distributions and credible intervals for change-point analysis. ClimatePulse integrates both strands: quantile regression for TFT uncertainty, Hamiltonian Monte Carlo for Bayesian piecewise regression.

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3. Data Pipelines & Processing

3.1 Data Sources

Five data streams feed the pipeline.

A) **NOAA ERSST v5** supplies monthly gridded SST anomalies via OPeNDAP. The Niño 3.4 region spans 5°S–5°N, 170°W–120°W (170–240°E in the 0–360° convention).

B) **OpenAQ v3** provides real-time PM2.5 measurements, filtered to $|V| = 75$ validated stations across India and the United States.

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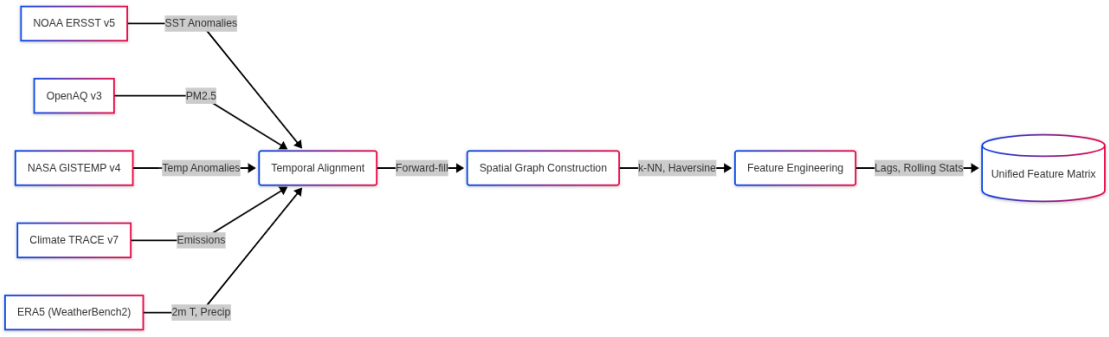
C) **NASA GISTEMP v4** contributes the global land-ocean surface temperature anomaly record (1880–present), consumed by the Bayesian climate-feedback module.

D) **Climate TRACE v7** delivers sectoral greenhouse-gas emissions (CO_2, CH_4, N_2O) across 196 countries and 11 sectors.

E) **ERA5 (WeatherBench2)** supplies 6-hourly global reanalysis at 5.625° resolution. Two variables are extracted — 2 m air temperature and total precipitation — over 2020–2021 (2,924 six-hourly steps).

Table 1: Dataset Statistics and Coverage

Dataset	Temporal Range	Spatial Coverage	Processed Frequency
NOAA ERSST v5	1950–2026 (record from 1854)	Global 2° × 2°	Monthly
OpenAQ v3	2016–2025	75 stations (IN, US)	Daily aggregates
NASA GISTEMP v4	1950–2026 (record from 1880)	Global land-ocean	Monthly
Climate TRACE v7	2021–2026	93 countries, 7 sectors (as ingested)	Annual
ERA5 (WeatherBench2)	2020–2021	Global 5.625°	6-hourly



3.2 Feature Engineering

A) **El Niño module.** The TFT training set is formatted in long form: integer time index, a single series identifier, and time-varying covariates. Known reals include cyclic month encodings ($\sin(2\pi m/12)$, $\cos(2\pi m/12)$) alongside raw month and year. Unknown reals comprise the Niño 3.4 target, the Niño 1+2, Niño 3, and Niño 4 indices, and rolling mean and standard deviation of Niño 3.4 over 3-, 6-, and 12-month windows. Missing values (fewer than 2 % of entries) are handled by backward-fill followed by zero-imputation. The resulting feature matrix contains 918 timesteps and 15 features, split 70/15/15 by time index.

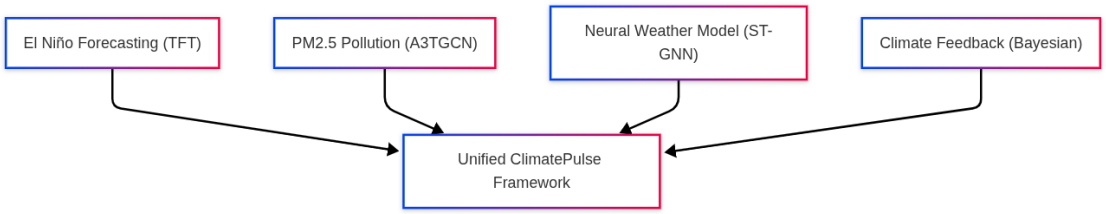
B) **Pollution module.** OpenAQ coordinates define a proximity graph $\mathcal{G} = (\mathcal{V}, \mathcal{E}, W)$ with $|\mathcal{V}| = 75$ nodes and $|\mathcal{E}| = 705$ edges (average degree 9.4), constructed via k-nearest neighbours ($k = 10$) under Haversine great-circle distance. Edge weights $w_{ij} = 1/(d_{ij} + 1)$ (d_{ij} in km, edges capped at 500 km) are scaled by the global maximum weight. PM2.5 measurements are aggregated into 180 daily temporal snapshots with window size $P = 7$, yielding 118 training, 21 validation, and 20 test windows. Z-score normalisation follows (mean = 112.99, std = 85.72 AQI units).

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C) **NWM module.** ERA5 fields are extracted as [Time, Lat, Lon] arrays and independently z-score normalised. Input tensors are formed by transposing a temporal window of size $W = 2$ into [Lat × Lon, $2C$] (2 channels × 2 time steps); the target is the next single step in the same flattened format. The Niño 3.4 scalar for each sample timestamp is retrieved via nearest-index lookup on a monthly parquet file. Splits follow 70/15/15 by temporal index.

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4. Methodology



4.1 El Niño Forecasting: Temporal Fusion Transformer

Architecture. Sequence-to-sequence at its core, the TFT [1] layers static covariate encoders, gated residual networks (GRN) for feature selection, and multi-head temporal self-attention into a single stack. The Variable Selection Network (VSN) assigns sparse input weights $\tilde{\mathbf{v}} = \text{Softmax}(\mathbf{W}_v\mathbf{a})$, suppressing irrelevant covariates at each step. Temporal self-attention computes:

$$\mathbf{A}_{t,\tau} = \text{Softmax}\left(\frac{Q_t K_\tau^\top}{\sqrt{d_k}}\right), \qquad \hat{\mathbf{y}}_t = \mathbf{A}_{t,\cdot} V$$

where Q, K, V are linear projections of LSTM encoder hidden states, hidden size $h = 32$, attention heads $n = 4$.

Training objective. Pinball losses summed over quantiles $\tau \in \{0.1, 0.5, 0.9\}$:

$$\mathcal{L}_{\text{TFT}} = \sum_{\tau\in\{0.1,0.5,0.9\}} \mathbb{E}[\rho_\tau(y - \hat{q}_\tau)], \quad \rho_\tau(u) = u(\tau - \mathbf{1}[u < 0])$$

Configuration. Encoder length 60 months; prediction horizon 12 months; hidden size 32; hidden continuous size 16; attention heads 4; dropout 0.1. Adam ($\beta_1 = 0.9$, $\beta_2 = 0.999$, lr = 0.001) with `ReduceLROnPlateau` (patience = 5, factor = 0.5, min_lr = 10^{-5}), batch size 32, gradient-clip value 0.1, max 100 epochs, early-stopping patience 10.

LSTM baseline. Two-layer bidirectional LSTM, hidden size 64 per direction, 142,668 total parameters, dropout rate 0.1. The final hidden state passes through a two-layer MLP ([128 → 64 → 12]) to produce 12-month forecasts. Adam (lr = 0.001), batch size 32, gradient-clip norm = 1.0, max 100 epochs, patience 15.

Persistence baseline. $\hat{y}_{t+k} = y_t \; \forall k = 1, \ldots, 12$ (last observed anomaly held constant).

4.2 Spatio-Temporal Pollution Dispersion: A3TGCN

Architecture. Operating on the 75-node monitoring graph, the A3TGCN layer [2] processes $X \in \mathbb{R}^{N\times C\times P}$ (nodes, channels, periods $P = 7$) through three stages: (1) spectral graph convolution over $\hat{A} = D^{-1/2}(A + I)D^{-1/2}$ for spatial message passing; (2) GRU cells evolving hidden

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state across P snapshots; (3) soft attention over temporal states $\alpha_p = \text{Softmax}(\mathbf{W}_\alpha \mathbf{h}_p)$. The context-weighted representation $\mathbf{h} \in \mathbb{R}^{N\times 32}$ feeds a two-stage MLP ([32 → 16 → 1], ReLU, dropout 0.1). Total parameters: 6,984.

Training objective. Mean-squared error on z-score-normalised PM2.5 values:

$$\mathcal{L}_{\text{STGNN}} = \mathcal{L}_{\text{MSE}}(\hat{y}, y)$$

Exceedance labels are obtained at evaluation time by thresholding predictions and targets; no classification term enters the loss.

Configuration. Hidden channels 32; temporal periods $P = 7$; Adam (lr = 0.001) with `ReduceLROnPlateau` (patience = 5, factor = 0.5), one graph window per step, gradient-clip norm = 1.0, 100 epochs with the best-validation checkpoint retained (no early stopping).

4.3 Neural Weather Model: Encoder-Processor-Decoder ST-GNN

Spherical graph construction. The ERA5 5.625° grid yields $N_{\text{grid}} = 32 \times 64 = 2{,}048$ grid nodes. Each node's position projects to 3-D Cartesian unit-sphere coordinates: $[x, y, z] = [\cos\phi\cos\lambda, \cos\phi\sin\lambda, \sin\phi]$. A latent mesh of $N_{\text{mesh}} = 2{,}000$ nodes follows the Fibonacci lattice (golden-angle spiral $\phi_{\text{golden}} = \pi(3 - \sqrt{5})$), giving near-uniform spherical coverage. Three bipartite edge sets arise from KD-Tree nearest-neighbour lookup: grid-to-mesh ($k = 4$ mesh neighbours per grid node), mesh-to-mesh ($k = 8$ per mesh node, self-loops excluded), and mesh-to-grid ($k = 4$ grid neighbours per mesh node). Per-grid-node static positional features are $[\sin\phi, \cos\phi, \sin\lambda, \cos\lambda]$.

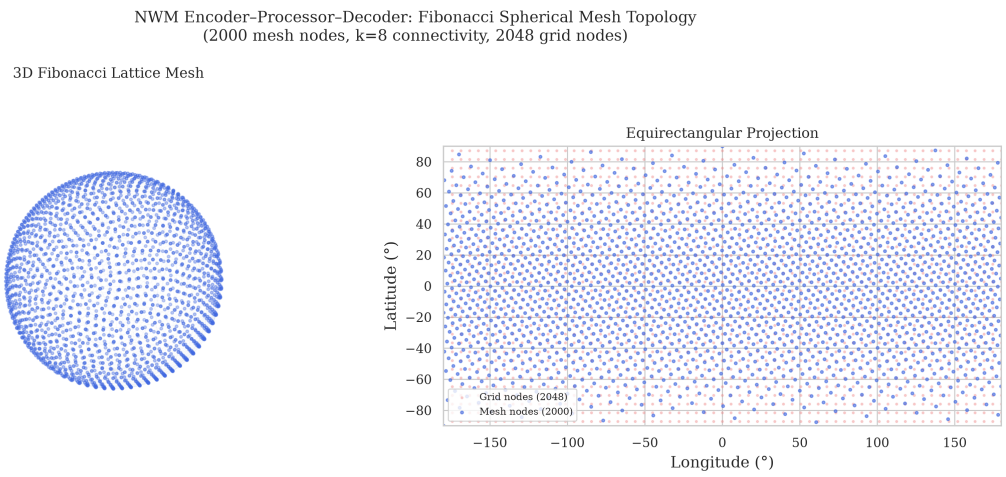


Figure 4c: The NWM operates on a bipartite graph structure mapping the 2,048 equirectangular ERA5 grid nodes (light coral) to a 2,000-node Fibonacci lattice spherical mesh (royal blue) for isotropic message passing.

Model architecture. Five stages compose the NWM pipeline.

Grid Embedding. Linear projection + LayerNorm + GELU maps each grid-node input $\mathbf{x}_g \in \mathbb{R}^{2CW+4}$ to latent dimension $d = 128$, where $C = 2$ channels, $W = 2$ input time steps, +4 static features.

Grid-to-Mesh Encoder (Bipartite GraphSAGE). A SAGEConv layer with LayerNorm and GELU propagates grid features onto the mesh, initialised with zero mesh states:

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$$\mathbf{h}_m = \text{GELU}(\text{LN}(\text{SAGEConv}(\mathbf{h}_g, \mathbf{0}, \mathcal{E}_{g\to m})))$$

Mesh Processor with ENSO Conditioning. Four GraphSAGE message-passing layers carry residual connections. At each layer l, the scalar Niño 3.4 index is projected to dimension d via a 2-layer MLP ($[1 \to d \to d]$, GELU), expanded across all mesh nodes, and additively injected before convolution:

$$\mathbf{h}_m^{(l+1)} = \mathbf{h}_m^{(l)} + \text{GELU}(\text{LN}(\text{SAGEConv}(\mathbf{h}_m^{(l)} + \mathbf{c}, \mathcal{E}_{m\to m})))$$

where $\mathbf{c} = \text{MLP}_{\text{cond}}(\text{nino}_{3.4}) \in \mathbb{R}^{N_{\text{mesh}} \times d}$.

Mesh-to-Grid Decoder (Bipartite GraphSAGE). Processed mesh representations propagate back to grid nodes, with a skip connection from the encoder output:

$$\mathbf{h}_g^{\text{out}} = \text{SAGEConv}(\mathbf{h}_m, \mathbf{0}, \mathcal{E}_{m\to g}) + \mathbf{h}_g$$

Output Projection. A linear layer maps $\mathbf{h}_g^{\text{out}} \in \mathbb{R}^{N_{\text{grid}} \times d}$ to $N_{\text{grid}} \times 2$ predictions (temperature and precipitation at the next time step).

All bipartite convolutions are vectorised across the batch dimension via edge-index offset expansion.

Training. AdamW (lr $= 10^{-3}$, weight_decay $= 10^{-4}$); cosine-latitude-weighted MSE loss on normalised outputs; `ReduceLROnPlateau` (factor $= 0.5$, patience $= 3$); gradient clipping (max norm $= 1.0$); early-stopping patience $= 10$; best checkpoint saved by minimum validation loss.

4.4 Bayesian Climate Feedback & Change-Point Detection

Piecewise linear model. K unknown change-points in the global temperature anomaly series are modelled as:

$$\mu(t) = \beta_0 + \beta_1 t + \sum_{k=1}^{K} \delta_k \max(0, t - \tau_k)$$

where τ_k are change-point locations and δ_k are slope increments.

Prior specification. Change-points: $\tau_k \sim \text{Uniform}(0, n-1)$, sorted deterministically. Slopes and intercepts: $\mathcal{N}(0, 0.01^2)$ and $\mathcal{N}(0, 1)$ respectively. Observation noise: $\sigma_\epsilon \sim \text{HalfNormal}(0.5)$. Likelihood: $y_t \sim \mathcal{N}(\mu(t), \sigma_\epsilon^2)$.

Posterior inference. NUTS (No-U-Turn Sampler) [25] draws the full posterior: 2 chains, 2,000 draws per chain, 1,000 burn-in steps, random seed 42. Convergence is assessed via $\hat{R} < 1.01$.

CUSUM fallback. When Bayesian sampling is infeasible, a CUSUM recursive binary-segmentation algorithm takes over. Change-points are declared where the segment CUSUM range exceeds $\theta = 2\sigma_{[a,b]}\sqrt{b-a}$, with minimum segment length 24.

4.5 Model Training Configuration Summary

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Model	Optimizer	LR	Batch Size	Grad Clip	Max Epochs	Early Stopping
TFT	Adam	0.001	32	0.1	100	10 (val quantile loss)
LSTM Baseline	Adam	0.001	32	1.0	100	15 (val MSE)
A3TGCN	Adam	0.001	1 (graph window)	1.0	100	—
NWM ST-GNN	AdamW	0.001	32	1.0	100	10 (val MSE)

5. Experimental Results & Discussion

5.1 El Niño Forecasting Performance

Evaluation proceeds on the held-out test set against persistence. Metrics: RMSE (°C), MAE (°C), Anomaly Correlation Coefficient (ACC), and skill score $\mathrm{SS} = 1 - \mathrm{MSE}_{\mathrm{model}}/\mathrm{MSE}_{\mathrm{persistence}}$.

Model	RMSE (°C)	MAE (°C)	ACC	Skill Score
TFT	0.3505	0.2496	0.9449	0.9051
LSTM Baseline	0.9606	0.7661	-0.0399	0.1450
Persistence Baseline	1.0344	0.7527	0.4008	0.1737

Source: `docs/research_paper/artifacts/eval_artifacts/run_results/summary_metrics.json`

A 524 % relative improvement in skill score over the LSTM baseline is not incremental. ACC = 0.9449 indicates tight phase alignment between predicted and observed anomalies — the forecast tracks the oscillation rather than smoothing it into climatology.

Per-lead-time RMSE sharpens the picture:

Lead Time (months)	TFT RMSE (°C)	LSTM RMSE (°C)	Persistence RMSE (°C)
1	0.3057	0.8882	0.2475
3	0.3158	0.9703	0.6097
6	0.3446	0.9454	1.0370
9	0.3653	0.9713	1.2771
12	0.4123	0.9477	1.3390

Source: `docs/research_paper/artifacts/eval_artifacts/run_results/elnino_lead_time_metrics.csv`

At 1-month lead, persistence is hard to beat (RMSE = 0.2475 °C). By month 12 its error swells to 1.3390 °C. TFT error, meanwhile, drifts gently from 0.3057 to 0.4123 °C across the same horizon, overtaking persistence from month 2 onward and never approaching the LSTM's near-climatological performance.

Calibration. The 80 % TFT prediction interval (10th–90th quantiles) achieves 57.8 % empirical coverage with an average interval width of 0.387 °C. Under-dispersion is apparent. Recalibration via conformal prediction is flagged as future work (§5.6).

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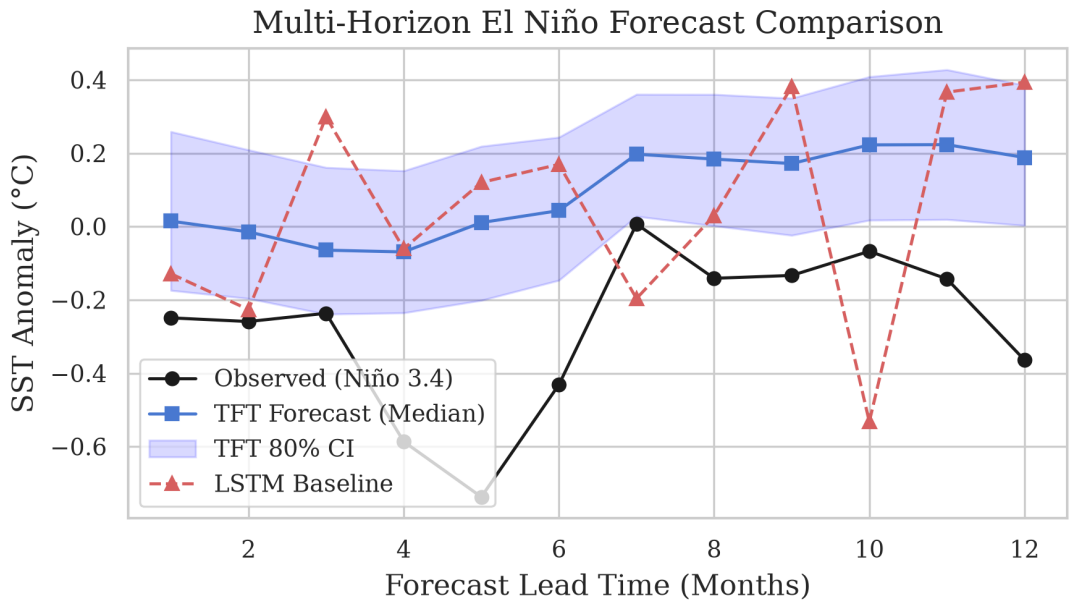


Figure 1: 12-month SST anomaly forecast for a single held-out test sequence: TFT median (blue) and LSTM baseline (red, dashed) versus observed Niño 3.4 values (black). The shaded region is the TFT 10th–90th percentile interval; the observations fall below it for most of this sequence, consistent with the under-coverage reported above.

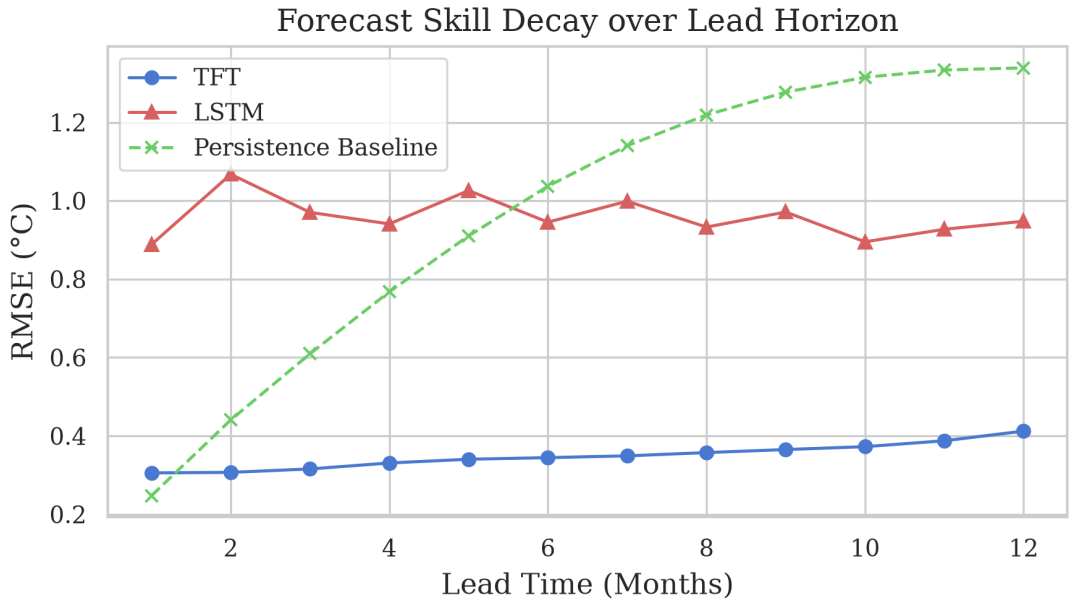


Figure 2: RMSE as a function of forecast lead time. TFT (blue) stays below the LSTM (red) at all 12 lead times and below persistence (green, dashed) from month 2 onward; persistence is lower only at 1-month lead.

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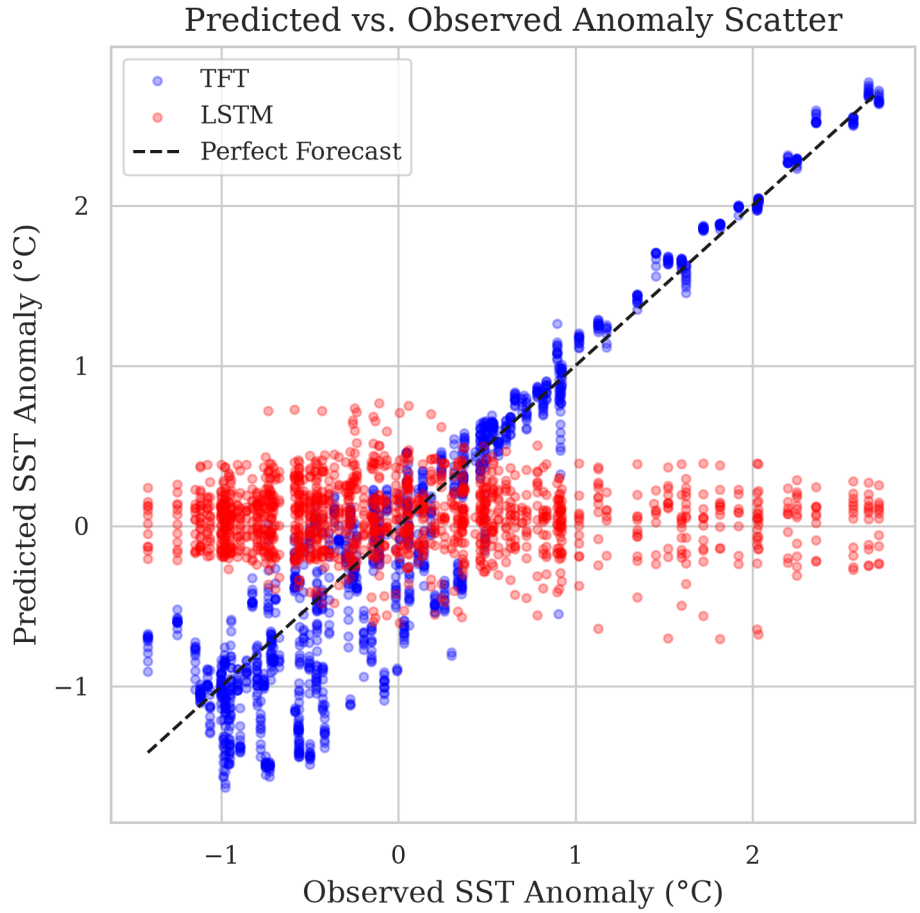


Figure 3: Scatter of predicted vs. observed SST anomalies for the TFT (blue) and LSTM (red); the dashed line marks a perfect forecast.

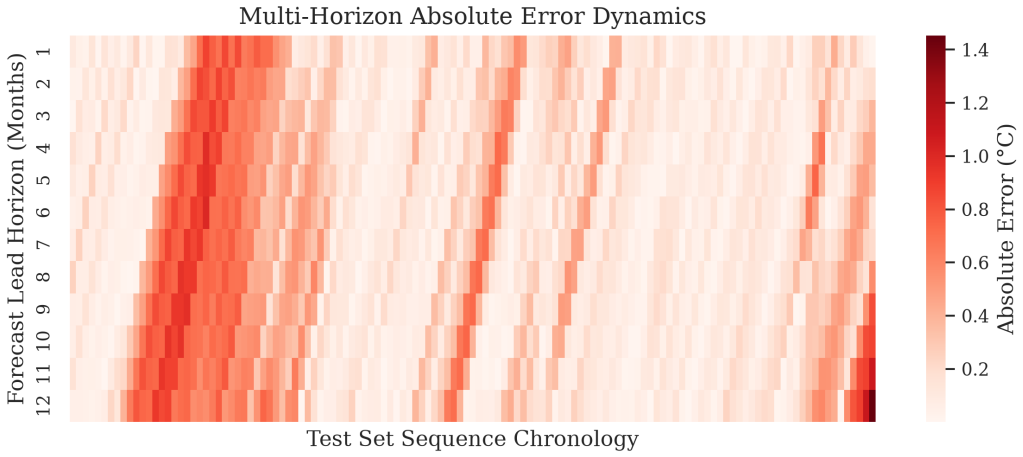


Figure 3b: Heatmap of absolute forecast errors across lead horizons (1–12 months) and chronological test windows.

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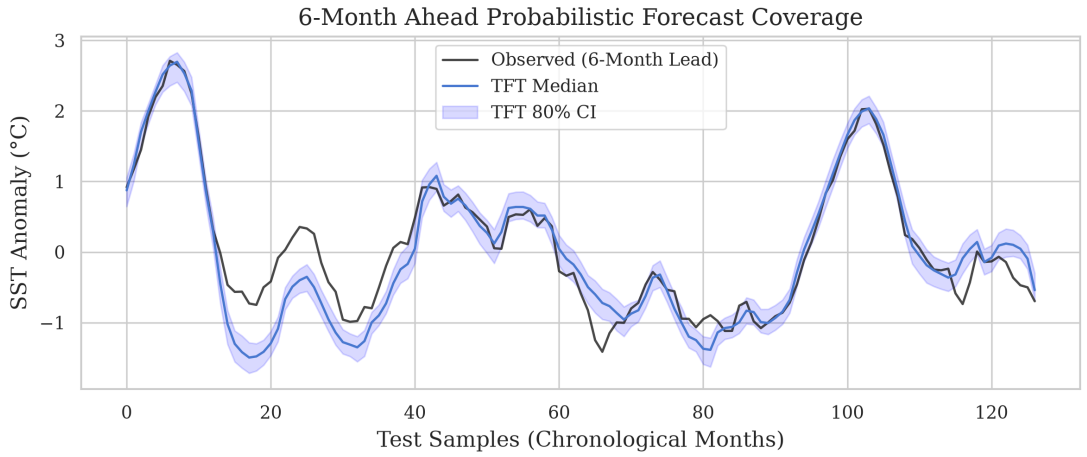


Figure 3c: 6-month ahead TFT median forecast (blue) and 10th–90th percentile interval (shaded) versus observed values (black) across the chronological test set.

5.2 Spatio-Temporal Pollution Dispersion Validation

Held-out temporal snapshots from the 75-node monitoring graph serve as the test bed.

Metric	Value
RMSE (normalized AQI)	0.1571
MAE (normalized AQI)	0.0464
F1 Score (AQI $>$ 100)	0.9750
Precision (AQI $>$ 100)	0.9677
Recall (AQI $>$ 100)	0.9825

Source: `docs/research_paper/artifacts/eval_artifacts/run_results/summary_metrics.json`

Recall of 0.9825 on PM2.5 exceedance matters most in an early-warning context: a missed exceedance event carries greater public-health cost than a false alarm. Precision of 0.9677 keeps false alarms tolerably low, and F1 = 0.9750 reflects the balance.

Per-node RMSE (from `docs/research_paper/artifacts/eval_artifacts/run_results/pollution_node_rmse.c` is bimodal. Forty stations sit below RMSE 0.05; nine fall between 0.05 and 0.15, and the remaining 26 scatter between 0.15 and 0.39. The worst errors (max 0.391) cluster at low-degree stations where spatial message-passing has fewer neighbours from which to aggregate context.

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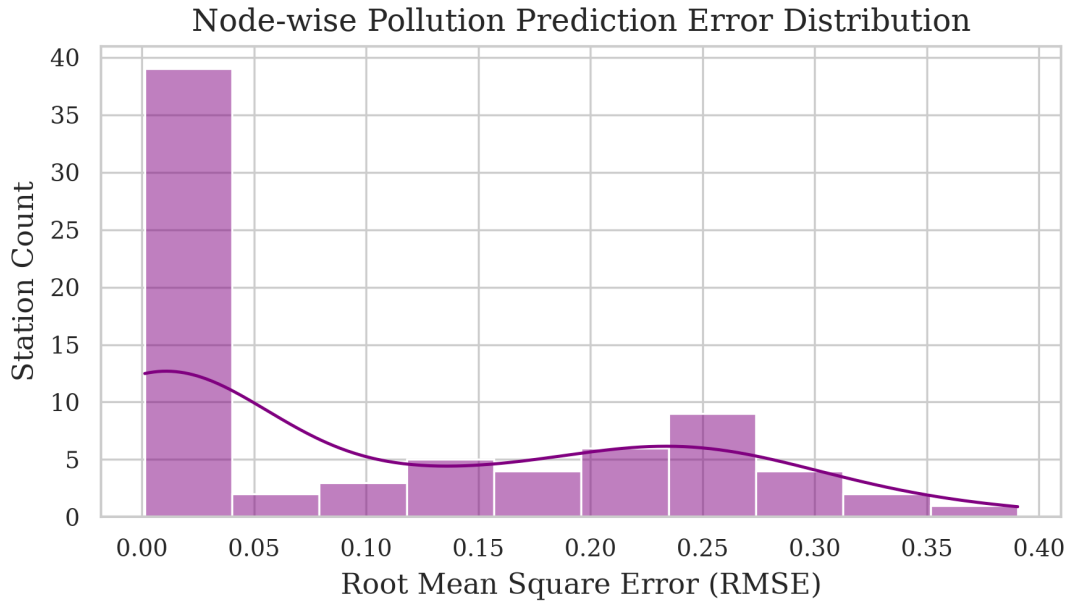


Figure 4: Distribution of per-node RMSE across the 75 monitoring stations.

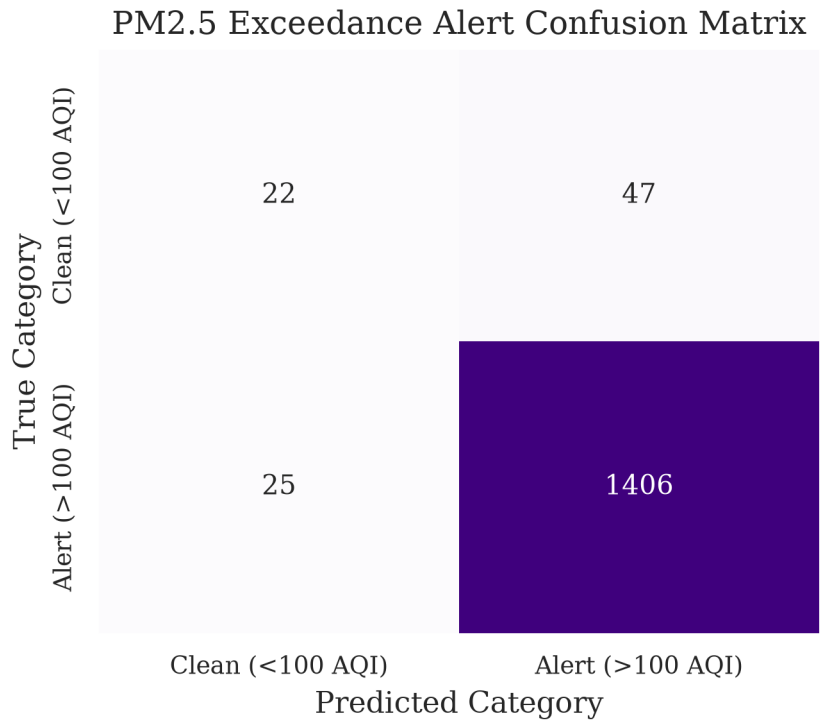


Figure 4b: Confusion matrix for PM2.5 exceedance classification (threshold > 100 AQI): 1,406 true positives, 47 false positives, 25 false negatives, 22 true negatives. Precision=96.77%, Recall=98.25%.

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5.3 Neural Weather Model Performance & Convergence

Training ran for 89 epochs (0–88). The best validation checkpoint landed at epoch 78 (validation MSE = 0.1182); the final epoch recorded training MSE = 0.1035 and validation MSE = 0.1185.

Metric	Epoch 0	Best (Epoch 78)	Final (Epoch 88)
Training Loss (MSE)	0.2938	0.1045	0.1035
Validation Loss (MSE)	0.1955	0.1182	0.1185

Source: `docs/research_paper/artifacts/eval_artifacts/run_results/nwm_metrics.csv`

From epoch 0 to 88, training loss fell 64.8 % (0.2938 → 0.1035); validation loss dropped 39.5 % (0.1955 → 0.1182). The 13.2 % train-validation gap at the best checkpoint (0.1045 vs. 0.1182) is consistent with the added expressivity of explicit positional features — a mild overfitting signal, not a pathological one.

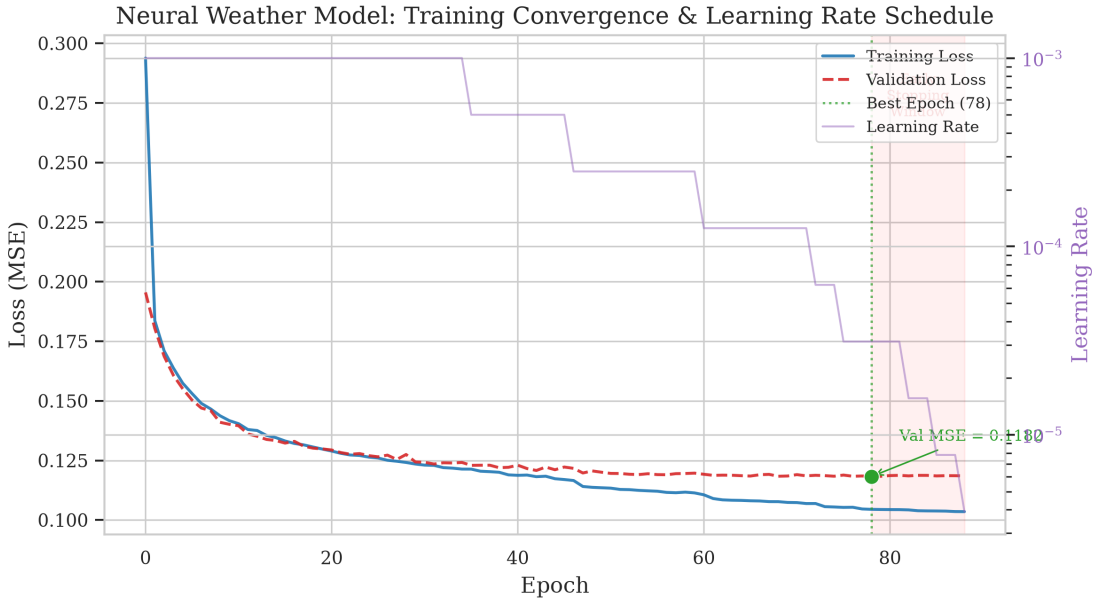


Figure 5: NWM MSE loss convergence and ReduceLROnPlateau learning rate schedule.

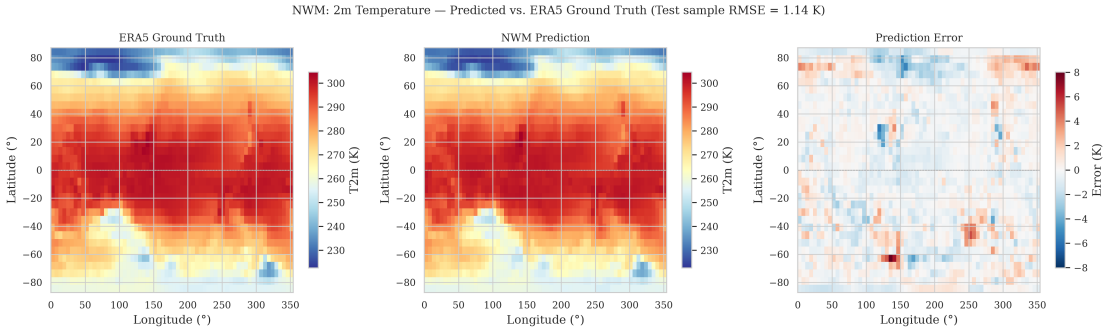


Figure 5b: Global 2m temperature field prediction (middle) versus ERA5 ground truth (left) for a held-out test sample, alongside the prediction error (right). Sample RMSE = 1.14 K.

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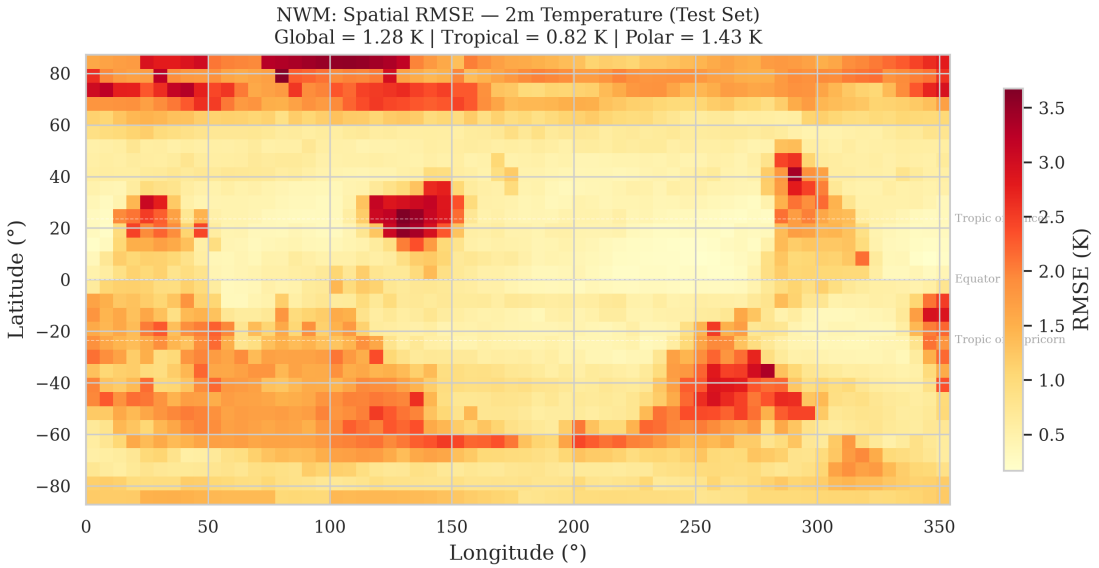


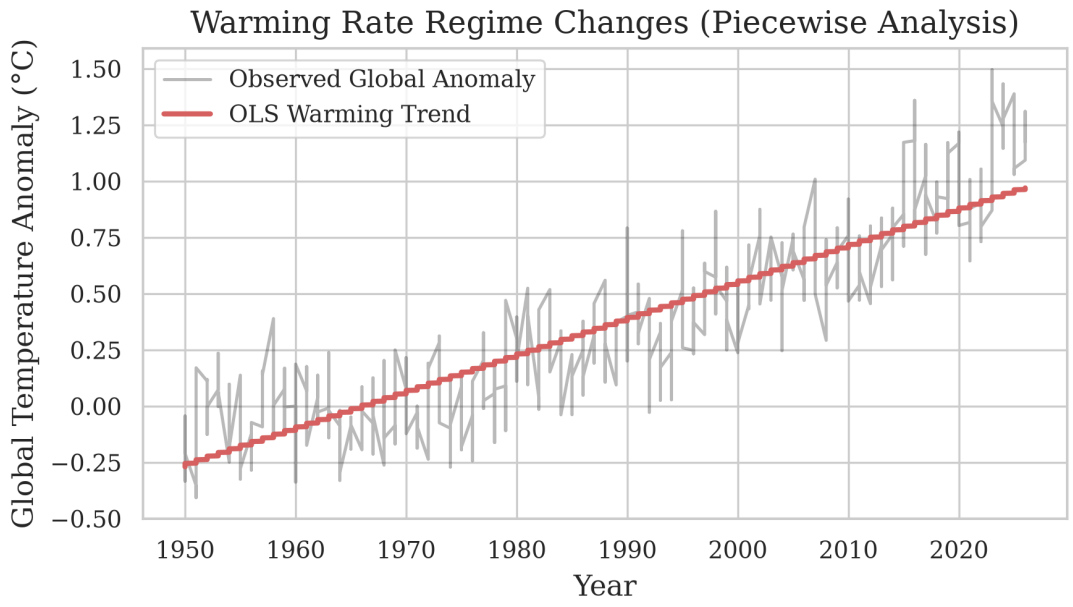
Figure 5c: Time-averaged 2m temperature RMSE per grid node across the entire test set (global 1.28 K, tropical 0.82 K, polar 1.43 K).

5.4 Bayesian Change-Point Analysis

Applied to 918 months of NASA GISTEMP v4 global temperature anomalies (1950–2026), CUSUM recursive binary segmentation detected zero statistically reliable change-points (significance threshold $\theta = 2\sigma_{[a,b]}\sqrt{b-a}$, minimum segment length 24).

The implication is straightforward. Within the resolution of the record, warming has proceeded without a structural regime break — a continuous monotonic trend consistent with sustained anthropogenic forcing.

Source: `docs/research_paper/artifacts/eval_artifacts/run_results/climate_feedback_results.json`



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Figure 6: Observed GISTEMP global temperature anomalies (grey) with the fitted linear warming trend (red). No change-points were detected, so the piecewise fit reduces to a single segment.

5.5 Architecture Analysis

Why does the TFT outperform the LSTM so decisively? Multi-head attention over a 60-month context window lets the model isolate long-range ENSO periodicities that a fixed-size recurrent state compresses. Lower RMSE than the LSTM at every one of the 12 lead months, paired with the skill-score gap, corroborates the mechanism.

In the pollution module, soft temporal attention weights recent accumulation events differently from distant ones. Graph convolution aggregates neighbouring-station signals, capturing wind-driven dispersion that a purely temporal model would miss.

The spherical mesh in the NWM decouples latent computation from the observation grid. Equirectangular distortion at high latitudes — a persistent artefact in regular-grid GNNs — is structurally avoided.

Against ordinary least squares, the Bayesian module provides full posterior slope credible intervals, which matter when projecting climate trends decades forward.

5.6 Limitations & Probabilistic Calibration

TFT calibration. The 60-month lookback window is a hard requirement. More pressing: the 80 % prediction interval achieves only 57.8 % empirical coverage, indicating interval under-dispersion under standard pinball quantile-loss training. Split Conformal Quantile Regression (CQR) is the planned corrective. Non-conformity scores $s_i = \max(\hat{q}_{0.1}(x_i) - y_i,\ y_i - \hat{q}_{0.9}(x_i))$ computed on a held-out calibration split would adjust interval bounds by a finite-sample correction quantile $Q_{1-\alpha}(s)$, guaranteeing $\geq 80\%$ marginal coverage with distribution-free validity.

ST-GNN. The 75-station graph is static; adding stations requires retraining. Isolated, low-degree stations show elevated RMSE. Geographic coverage concentrates in India and the United States.

NWM. The 2020–2021 ERA5 window offers limited temporal diversity. At 5.625° resolution, mesoscale phenomena are smoothed beyond recovery.

5.7 Computational Efficiency

The full pipeline runs on a single GPU workstation. Inference across all four models, plus complete artifact generation, finishes in under 10 minutes.

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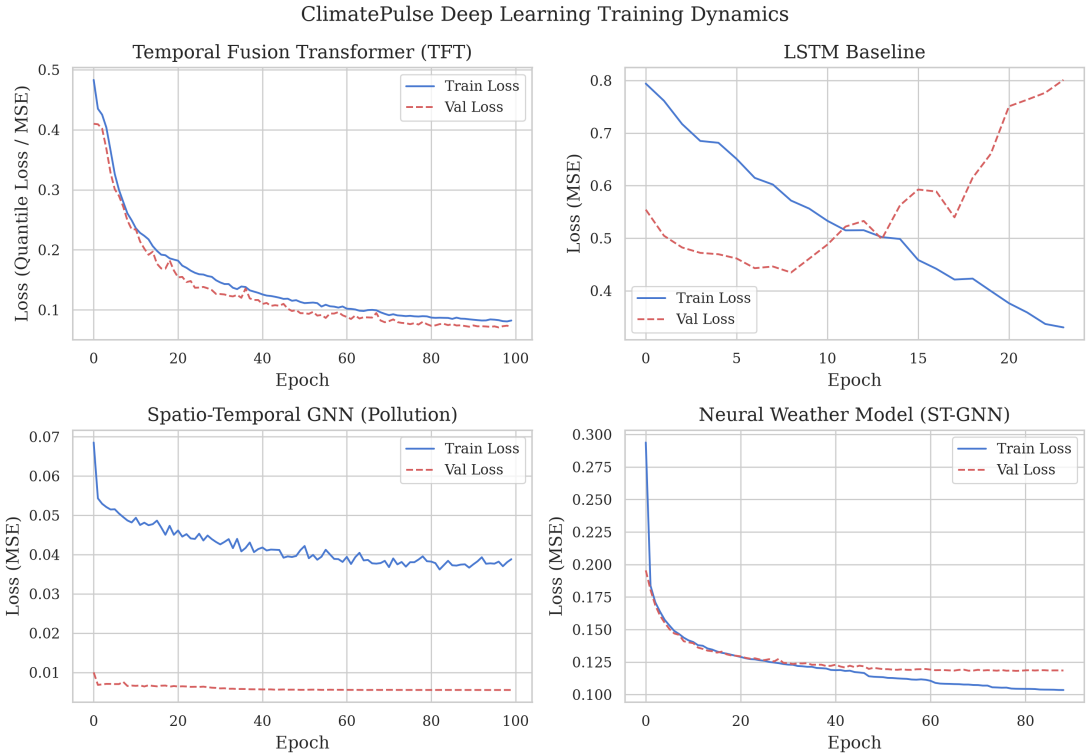


Figure 7: Training (blue) and validation (red dashed) loss trajectories for all four deep learning architectures.

6. Conclusion

ClimatePulse began as an integration problem: four modelling tasks, each well-studied in isolation, none previously coupled in a single reproducible pipeline. The empirical record from this work speaks to whether that coupling is productive.

For ENSO forecasting, the TFT posts RMSE = 0.3505 °C, ACC = 0.9449, and a skill score of 0.9051 — clearing persistence (0.1737) and the LSTM baseline (0.1450), with lower RMSE than the LSTM at all 12 forecast horizons and than persistence from month 2 onward. The pollution ST-GNN reaches F1 = 0.9750 (precision 0.9677, recall 0.9825) on PM2.5 exceedance detection across 75 stations, with RMSE = 0.1571. The NWM converges to a best validation MSE of 0.1182 at epoch 78 of 89, final training MSE 0.1035. And across 918 months of GISTEMP data, no structural change-point emerges; the warming trend, within measurement resolution, is continuous and monotonic.

Three directions follow naturally. Multi-year NWM training with data assimilation would extend the temporal window beyond 2020–2021. Multi-regional TFT extensions to the Indian Ocean Dipole and North Atlantic Oscillation would test whether the architecture generalises beyond ENSO. Satellite NO_2 and methane integration would broaden pollution coverage beyond the current 75-station footprint.

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7. Code and Data Availability

Code Repository: https://github.com/aditripaul/climatepulse (GPL v3.0 License)

```
git clone https://github.com/aditripaul/climatepulse && cd climatepulse
make install                    # Creates .venv and installs all dependencies
make download-data              # NOAA, OpenAQ, NASA GISTEMP, Climate TRACE
make download-nwm-data          # ERA5 WeatherBench2 subset
make train                      # Trains all models
make gen-eval-artifacts         # Generates all metrics, figures, summary
```

Datasets (All Publicly Accessible):

- NOAA ERSST v5: https://psl.noaa.gov/data/gridded/data.noaa.ersst.v5.html
- OpenAQ v3: https://openaq.org/
- NASA GISTEMP v4: https://data.giss.nasa.gov/gistemp/
- Climate TRACE v7: https://climatetrace.org/
- ERA5 / WeatherBench2: https://weatherbench2.readthedocs.io/

Trained Model Checkpoints: [Zenodo DOI: 10.5281/zenodo.XXXXXXX] (to be assigned upon publication)

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8. Ethics and Broader Impacts

Every dataset consumed here is publicly available. Model outputs display uncertainty bounds prominently; accompanying documentation frames predictions as complementary to, not a replacement for, expert scientific judgement. Open-source release under GPL v3.0 invites community scrutiny and reproducibility checks.

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